



DEPARTMENT OF PHYSICS PROFILE

❖ About the Department:

The Department of Physics has been committed to excellence in teaching and learning since its inception. The department aims to nurture scientific curiosity, analytical thinking, and a deep understanding of fundamental and applied physics among undergraduate students. With a student-centric approach and a well-qualified faculty, the department prepares students for higher studies, research, and diverse career paths.

❖ Vision:

To be a center of excellence in physics education that fosters innovation, research, and societal development.

❖ Mission:

- To provide quality education in physics through effective teaching-learning practices.
- To instill scientific temper and critical thinking in students.
- To promote hands-on experience through laboratory training.
- To prepare students for academic, industrial, and research careers.

❖ Objectives:

- To provide a strong foundation in theoretical and experimental physics.
- To align the curriculum with the latest developments in physics and related interdisciplinary fields.
- To develop students' analytical thinking, problem-solving, and computational skills.
- To encourage undergraduate research and participation in conferences, workshops, and science fairs.
- To prepare students for higher education in physics and allied disciplines.

- To provide guidance for competitive examinations and career opportunities in academia, research, and industry.
- To foster curiosity, creativity, and critical thinking among students.
- To popularize physics through outreach programs, exhibitions, and lectures for school and community audiences.

❖ Academic Programs:

- **B.Sc. MPC** (Mathematics, Physics and Chemistry)
- **B.Sc. MPCs** (Mathematics, Physics and Computer Science)

❖ Career Opportunities for Graduates:

- Higher studies (M.Sc.),
- Competitive exams (CPGET, IIT-JAM, UPSC, SSC, etc.)
- Employment in teaching, IT, electronics, scientific labs, and public sector.

❖ Faculty:

S. No.	Name of the faculty	Period of Work
1	Hemalatha, B.Ed., M.Sc.	From 2022 to 2024 July
2	Dr. Chittimadula Madhuri, M.Sc., Ph.D.	From July 2024 to Till Date
3	D. Sweta B.Ed., M.Sc.	From July 2024 to March 2025

+ Dr. Chittimadula Madhuri, M.Sc., TS-SET, Ph.D.

+ Educational Qualifications:

- ✓ She has been awarded Ph.D. Physics from the Osmania University in the year 2022 for the topic “Investigation of ceria based nanocomposite materials for low temperature solid oxide fuel cell applications” under the supervision of Prof. C. Vishnuvardhan Reddy, Osmania University.
- ✓ She has completed her M.Sc. Physics in 2013 from Department of Physics, University College of Science, Osmania University, Hyderabad.
- ✓ She has qualified State Eligibility Test (TS-SET)–2018 and GATE–2015.



Teaching Experience:

- ✓ She has total 05 Years of Teaching Experience.
- ✓ She worked as an Assistant Professor of Physics in Methodist College of Engineering and Technology, Abids, Hyderabad for 03 years from 2014 to 2017.
- ✓ Dr. Chittimadula Madhuri currently working as a Degree Lecturer in Physics since 2024 at MJPTBCWRDC, Station Ghanpur located at Pembarti, Jangaon.

Teaching Interests:

- Mechanics,
- Thermodynamics,
- Waves & Oscillations,
- Optics,
- Modern Physics,
- Electronics,
- Fundamentals of Nano Technology, and
- Engineering Physics.

Research Experience:

- ✓ She has total 08 Years of Research Experience.
- ✓ To her credit she had published Total of 17 papers in Peer reviewed International Journals indexed in SCI, published one Book Chapter and attended several conferences in India.
- ✓ Her areas of research are: Synthesis & characterization of Nano structured anode, cathode, and electrolyte as well as Nanocomposite materials for Solid Oxide Fuel Cells.

Awards / Achievements:

- ✓ **Best Oral Presentation** award for presenting “Enhanced Ionic Transport and Structural Properties of $\text{Ce}_{0.76}\text{Sm}_{0.08}\text{Nd}_{0.08}\text{Y}_{0.08}\text{O}_{2-8}$ tri-doped Ceria Electrolyte for IT-SOFC applications”, in the National Conference on “Emerging Materials for Energy and Environmental Applications (NCEMEEA-2026)” during 25–26 March 2026, organized by Government Degree College (Autonomous), Siddipet, Telangana, India.
- ✓ She has qualified State Eligibility Test (TS-SET)–2018 and GATE–2015.

Year wise student Intake and Admissions:

Year	Student Intake	Admission
2022-2023	80	10
2023-2024	80	22
2024-2025	80	28
2025-2026	80	29
2026-2027	80	37

Research Publications:

1. B. Swamy, B. Mallesham, Kasarapu Venkataramana, **Chittimadula Madhuri**, P. Rama Krishna, Ch Madhusudan, C. Vishnuvardhan Reddy, “Enhancing the ionic conductivity of ZrO_2 electrolyte via Sm^{3+} and Gd^{3+} Co-doping for intermediate temperature applications” **Ceramics International** 51(30) Part A (2025) 62323–62331.
2. Kasarapu Venkataramana, **Chittimadula Madhuri**, Ch. Madhusudan, C.Vishnuvardhan Reddy, Y.Ranjith Kumar, M. Vasundhara, “Structural, Micro-Structural, Ionic-Conductivity and Thermal Expansion Study of $\text{Ce}_{0.76}\text{Sm}_{0.08}\text{Nd}_{0.08}\text{Y}_{0.08}\text{O}_{2-\delta}$ Electrolyte for IT-SOFC Applications” **Journal of the Indian Chemical Society** (2025) 102122.
3. Kasarapu Venkataramana, **Chittimadula Madhuri**, Ch. Madhusudan, C.Vishnuvardhan Reddy, Y. Ranjith Kumar, M. Vasundhara, “A parametric study on triple doped ceria ($\text{Ce}_{0.76}\text{Pr}_{0.08}\text{Nd}_{0.08}\text{Gd}_{0.08}\text{O}_{2-\delta}$) electrolyte material for solid oxide fuel cells” **Ceramics International** 51 (2025) 15809–15817
4. B Swamy, Kasarapu Venkataramana, **Chittimadula Madhuri**, Ashok Bhogi, C. Vishnuvardhan Reddy, “Structural and Electrical Studies of Nd and Gd co-doped Zirconia Electrolytes for IT-SOFC applications” **Ionics** 30(4) (2024) 1–10
5. **Chittimadula Madhuri**, Kasarapu Venkataramana, S. Ramesh, Jada Shanker, C Vishnuvardhan Reddy, “Investigation of Triple-doped Ceria Based Composite Materials for LT-SOFC Applications” **Journal of Electronic Materials** 51 (2022) 5908–5918

6. **Kasarapu Venkataramana**, Chittimadula Madhuri, Ch Madhusudan, Ashok Bhogi, B. Srinivas, C Vishnuvardhan Reddy, “Influence of La^{3+} , Sm^{3+} and Dy^{3+} Dopants on Ceria Solid Electrolytes for IT-SOFCs” **Materials Sci. in Semiconductor Processing** 142 (2022) 106495
7. **Chittimadula Madhuri**, Kasarapu Venkataramana, J. Shanker, C. Vishnuvardhan Reddy “Effect of La^{3+} , Pr^{3+} , and Sm^{3+} triple-doping on structural, electrical, and thermal properties of ceria solid electrolytes for intermediate temperature solid oxide fuel cells” **Journal of Alloys and Compounds** 849 (2020) 156636.
8. Kasarapu Venkataramana, **Chittimadula Madhuri**, C. Vishnuvardhan Reddy “Triple-doped Ceria–Carbonate ($\text{Ce}_{0.82}\text{La}_{0.06}\text{Sm}_{0.06}\text{Gd}_{0.06}\text{O}_{2-\delta} - (\text{Li-Na})_2\text{CO}_3$) Nanocomposite Solid Electrolyte materials for LT-SOFC applications” **Ceramics International** 46 (2020) 27584–27594.
9. Kasarapu Venkataramana, **Chittimadula Madhuri**, C Madhusudan, Y. S Reddy, C. Vishnuvardhan Reddy “Investigation on micro-structural, structural, electrical and thermal properties of La^{3+} , Sm^{3+} & Gd^{3+} triple-doped ceria as solid-electrolyte for intermediate temperature-solid oxide fuel cell applications” **Journal of Applied Physics** 126 (2019) 144901.
10. Ch. Madhusudan, Venkataramana Kasarapu, **Madhuri Chittimadula**, C. Vishnuvardhan Reddy “Structural, electrical and thermal studies on microwave sintered Dy and Pr co-doped ceria ceramics as electrolytes for intermediate temperature solid oxide fuel cells” **J. Mater Sci: Materials in Electronics** 29 (2018) 17067–17077.
11. **Chittimadula Madhuri**, Kasarapu Venkataramana, A. Nurhayati, C. Vishnuvardhan Reddy “Effect of La^{3+} and Pr^{3+} co-doping on structural, thermal and electrical properties of ceria ceramics as solid electrolytes for IT-SOFC applications” **Current Applied Physics** 18 (2018) 1134–1142.
12. Ch. Madhusudan, Venkataramana Kasarapu, **Madhuri Chittimadula**, Y. S. Reddy, C. V. Reddy “Synthesis and characterization of Y and Dy co-doped ceria solid electrolytes for IT-SOFCs: a microwave sintering” **Rare Metals** 40 (2021) 3329–3336.
13. Kasarapu Venkataramana, **Chittimadula Madhuri**, C Madhusudan, Y Reddy, G. Bhikshamaiah, C. Vishnuvardhan Reddy “Investigation on La^{3+} and Dy^{3+} co-doped ceria ceramics with an optimized average atomic number of dopants for electrolytes in IT-SOFCs” **Ceramics International** 44 (2018) 6300–6310.

14. **Kasarapu Venkataramana**, Chittimadula Madhuri, J Shanker, Ch. Madhusudan, C. Vishnuvardhan Reddy “Microwave-sintered Pr^{3+} , Sm^{3+} , and Gd^{3+} triple-doped ceria electrolyte material for IT-SOFC applications” **Ionics** 24(10) (2018) 3075–3084.
15. Kasarapu Venkataramana, K Ravindranath, **Chittimadula Madhuri**, Ch. Madhusudan, N. Pavan Kumar, C. Vishnuvardhan Reddy “Low temperature microwave sintering of yttrium and samarium co-doped ceria solid electrolytes for IT-SOFCs” **Ionics** 24(5) (2018) 1429–1438.
16. Kasarapu Venkataramana, **Chittimadula Madhuri**, Y. S Reddy, G. Bhikshamaiah, C. Vishnuvardhan Reddy “Structural, electrical and thermal expansion studies of tri-doped ceria electrolyte materials for IT-SOFCs” **Journal of Alloys and Compounds** 719 (2017) 97–107.

Conference Proceedings:

17. K Ravindranath, Kasarapu Venkataramana, Chittimadula Madhuri, C. Vishnuvardhan Reddy “Preparation and characterization of Zirconium and samarium co-doped ceria solid electrolytes for IT-SOFC applications” *Materials Today: Proceedings*, 5 (13) (2018) 27142-27148.
18. Kasarapu Venkataramana, Chittimadula Madhuri, and C. Vishnuvardhan Reddy, “Enhanced ionic conductivity of co-doped ceria-carbonate nano composite electrolyte material for LT-SOFCs” *AIP Conference Proceedings* 1832, 050024 (2017); doi: 10.1063/1.4980257.
19. Kasarapu Venkataramana, Ch. Madhusudan, Chittimadula Madhuri and C. Vishnuvardhan Reddy, “Synthesis, Characterization and Thermal Expansion of $\text{Ce}_{0.8-x}\text{Sm}_{0.2}\text{Zr}_x\text{O}_{2-\delta}$ Solid Electrolytes for IT-SOFC applications” *Materials Today: Proceedings*, 3 (10) (2016) 3908-3913.

Book Chapters

1. Kasarapu Venkataramana, **Chittimadula Madhuri**, “Cerium based electrolyte materials for advanced solid oxide fuel cell (SOFC) applications”, *Materials for Advanced applications, IIP Series*, Vol.1, Chapter 10, 2024 pp.133-149. ISBN 978-93-6252-500-0.

List of papers presented in National/ International conferences

1. Enhanced Ionic Transport and Structural Properties of $\text{Ce}_{0.76}\text{Sm}_{0.08}\text{Nd}_{0.08}\text{Y}_{0.08}\text{O}_{2-\delta}$ tri-doped Ceria Electrolyte for IT-SOFC applications, in the National Conference on “Emerging Materials for Energy and Environmental Applications (NCMEMEA-2026)” during 25–26 March 2026, organized by Government Degree College (Autonomous), Siddipet, Telangana, India.

2. Synthesis and characterization of Tri-doped ceria Electrolyte for IT-SOFC Application, Telangana Science Congress-2025, 19-21 August 2025. Jointly Organized by Telangana Academy of Sciences and Kakatiya University, Warangal, Telangana.
3. Triple-Doped Ceria-Carbonate Nanocomposite Solid Electrolyte for Advanced SOFC Applications, International Conference on Recent Advances & Innovations in Technology, Management & Applied Sciences (ICRAITMS-2021), 19th and 20th March 2021, St. Martin's Engineering College, Dhulapally, Secunderabad, Telangana, India.
4. Preparation and characterization of $\text{Ce}_{0.8}\text{La}_{0.2-x}\text{Dy}_x\text{O}_{2-\delta}$ ceramics for solid electrolytes in intermediate temperature solid oxide fuel cells, international conference on advances in multifunctional materials (ICAFM-2018) 12th and 13th January 2018, KTHM College, Nasik, Maharashtra, India.

Academic Achievements:

Results:

A-Appeared; P-Pass; %-Pass Percentage

Batch	SEM-I			SEM-II			SEM-III			SEM-IV			SEM-V			SEM-VI		
	A	P	%	A	P	%	A	P	%	A	P	%	A	P	%	A	P	%
2022-2025	13	13	100	13	13	100	10	10	100	10	10	100	10	10	100	10	10	100
2023-2026	27	27	100	26	26	100	26	26	100	24	24	100	22	21	95.4	22	22	100
2024-2027	28	27	96	28	28	100	28	28	100	28	26	92.8						
2025-2028	27	26	96.2	27	26	96.2												

Course Objectives:

PAPER I: MECHANICS AND OSCILLATIONS:

CO1: Develop problem-solving, analytical skills and understand rotational motion and rigid-body dynamics

CO2: Apply central force concepts to gravitational systems. Perform Lorentz transformations and analyze relativistic kinematics

CO3: Understand the fundamental concepts of simple harmonic motion (SHM) and evaluate forced oscillations and resonance

CO4: Conceptual Understanding and to develop Mathematical and Analytical Skills

PAPER III: ELECTROMAGNETICS:

CO1: Understand Electrostatics

CO2: Apply Concepts of Magnetostatics

CO3: Understand Electromagnetic Induction and Analyze Time-Varying Fields and Maxwell's Equations

CO4: Understand AC Fundamentals and Apply Network Theorems to Electrical Circuits

PAPER V: MODERN PHYSICS:

CO1: Analyze Atomic Structure

CO2: Understand the Foundations of Modern Physics and Apply Quantum Mechanical Concepts

CO3: Understand Nuclear Physics

CO4: Understand crystal structures, symmetry and Apply diffraction principles to determine crystal structure.

PAPER II: THERMAL PHYSICS:

CO1: Understand Kinetic Theory of Gases and Apply Thermodynamic Principles

CO2: Understand Thermodynamic Potentials and Maxwell's Equations and Analyze Low Temperature Phenomena

CO3: Understand Quantum Theory of Radiation and Apply Radiation Laws to Thermal Measurements

CO4: Understand Statistical Mechanics and Analyze Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac Distributions

PAPER IV: WAVES AND OPTICS:

CO1: Understand Wave Motion and Vibrations

CO2: Understand Interference and Apply Interference Principles in Optical Systems

CO3: Understand Diffraction and Analyze Diffraction Patterns and Resolving Power

CO4: Understand Polarization and Apply the Principles of Polarized Light to Optical Instruments

PAPER VI: (A) ELECTRONICS (DSE-2: ELECTIVE):

CO1: Understand Semiconductor Physics and Analyze P-N Junction Diodes and Rectifiers

CO2: Understand Bipolar Junction Transistors and Apply Feedback and Oscillator Concepts

CO3: Understand Special Semiconductor Devices and Analyze Their Characteristics and Applications

CO4: Understand Digital Electronics and Apply Number Systems and Logic Gates

Programme Outcomes:

- To inculcate Fundamental Knowledge
- Enhance Practical and Experimental Skills
- To develop Analytical and Problem-Solving Abilities
- Application of Physics in Technology and Society
- To understand Research and Scientific Inquiry
- To develop Communication and Teamwork Skills

Academic Action plan

1. I-SEM: Mechanics and oscillations

- To provide students with a strong conceptual foundation in classical mechanics and oscillatory motion, including problem-solving and practical skills.
- To solve numerical problems and derivations on vector analysis, central forces, special theory of relativity, oscillations and waves.

2. III-SEM: Electromagnetic Theory

- To teach fundamentals of electricity, magnetism, and electromagnetic waves with problem-solving skills and practical application.
- To solve derivations of electrostatics, magnetostatics, electromagnetic waves, LCR circuits and network theorems, related problems.

3. V-SEM: Modern Physics

- To introduce students to quantum concepts, atomic/nuclear physics, and relativity, developing both conceptual understanding and problem-solving skills.
- Learning concepts of atomic and molecular spectroscopy, dual nature of matter waves, Schrodinger equations, nuclear physics, crystallography and bonding in crystals, related problems of all concepts.

4. II-SEM: Thermal Physics

- To develop a strong understanding of kinetic theory, thermodynamics, radiation, and statistical mechanics.
- To solve numerical problems and derivations related to Maxwell's distribution, thermodynamic laws, entropy, thermodynamic potentials, radiation laws, and statistical distributions.

5. IV SEM: Waves and Optics

- To develop a strong understanding of wave motion, interference, diffraction, and polarization.
- To solve numerical problems and derivations related to wave equations, interference, diffraction gratings, resolving power, polarization, and optical instruments.

6. VI SEM: Electronics

- To develop a strong understanding of semiconductor devices, transistors, amplifiers, oscillators, and digital electronics.
- To solve problems related to P-N junctions, rectifiers, Zener diodes, transistors, feedback, oscillators, special semiconductor devices, number systems, and logic gates.

Time table (A.Y.: 2025-2026)

	1	2	3	4	5	6	7
Mon	LAB: MPCs-II			MPCs-III		RC: MPCs-I	FNT
Tue	MPCs-II	MPCs-I		LAB: MPCs-III			
Wed			MPCs-II	RC: MPCs-II	MPCs-III	FNT	PG Coaching
Thrus		MPCs-I	MPCs-II	MPCs-III		LAB: MPCs-I	
Fri	MPCs-III	MPCs-II	MPCs-I			RC: MPCs-III	club 1
Sat			MPCs-I	Swachh Bharat			club 2

Syllabus of I, III, V semester

KAKATIYA UNIVERSITY-WARANGAL-TELANGANA

Under Graduate Courses (Under CBCS 2025–2026 onwards)

B.Sc (PHYSICS) - I Year, SEMESTER – I

Paper–I: Mechanics and Oscillations

w.e.f academic year (2025-26) (CBCS)

Total: 56 Hrs (4hrs/week)

UNIT–I

Vector Analysis: (7 hrs)

Scalar and Vector fields, Gradient of a Scalar field, Divergence and Curl of a Vector field and their physical significance and related problems. Vector integration, Line, Surface and Volume integrals. Applications of Stokes', Gauss's and Green's theorems.

Rigid body Dynamics: (7 hrs)

Definition of Rigid body, rotational kinematic relations, equation of motion for a rotating body, angular momentum and inertial tensor. Euler's equations, precession of a top, Gyroscope.

UNIT–II

Central Forces: (7 hrs)

Central forces-definition and examples, conservative nature of central forces, conservative force as a negative gradient of potential energy, equation of motion under a central force, gravitational potential and gravitational field, motion under inverse square law, derivation of Kepler's laws.

Special theory of Relativity: (7 hrs)

Galilean relativity, absolute frames, Michelson-Morley experiment, Postulates of special theory of relativity. Lorentz transformation equations, time dilation, length contraction, addition of velocities, mass-energy relation, Concept of four vector formalism.

UNIT – III

Oscillations: (14 hrs)

Simple harmonic oscillator and solution of the differential equation-Physical characteristics of

SHM, torsion pendulum measurements of rigidity modulus, compound pendulum, measurement of 'g', Damped harmonic oscillator, solution of the differential equation of damped oscillator, Energy considerations, logarithmic decrement, relaxation time, quality factor, differential equation of forced oscillator and its solution, amplitude resonance, velocity resonance.

UNIT–IV

Waves: (14 hrs)

Fundamentals of Waves -Transverse wave propagation along a stretched string, general solution of wave equation and its significance, modes of vibration of stretched string clamped at ends, overtones, energy transport, transverse impedance.

Longitudinal vibrations in bars- wave equation and its general solution,

Special cases: i) Bar fixed at both ends, ii) Bar fixed at the midpoint, iii)

Bar free at both ends, iv) Bar fixed at one end, Transverse vibrations in a

bar - wave equation and its general solution.


Chairperson
BOARD OF STUDIES
Department of Physics
Kakatiya University,
Warangal-505 009 (T.O.)

Note: Problems should be solved at the end of every chapter of all units.

Reference/Suggested Books

- 1) **Mechanics** by C.Kittel,W.Knight,M.A.Ruderman-Berkeley Physics Course.Vol.1, *Tata-Mc Grawhill Company Edition2008.*
- 2) **Fundamentals of Physics**.Halliday/Resnick/Walker *Wiley India Edition2007.*
- 3) **Theory of relativity - Resnick**
- 4) **First Year Physics-Telugu Academy, Telangana**
- 5) **Introduction to Physics for Scientists and Engineers. F.J. Ruche. McGraw Hill.**
- 6) **Fundamentals of Physics** by Alan Giambattistaetal *Tata-Mc Graw Hill Company Edition, 2008.*
- 7) **University Physics** byYoung and Freeman,*Pearson Education, Edition2005.*
- 8) **Sears and Zemansky's University Physics** by Hugh D.Young, Roger A.Freedman
Pearson Education Eleventh Edition.
- 9) **An introduction to Mechanics** by Daniel Kleppner & Robert Kolenkow. *The McGraw Hill Companies.*
- 10) **Mechanics** by Hans & Puri. *TMH Publications.*
- 11) **Engineering Physics. R.K.Gaur & S.L. Gupta. Dhanpat Rai Publications.**
- 12) **The Feynman Lectures in Physics, Vol.-1, R P Feynman, R B Lighton and MSands,**
BI Publications,
- 13) **Mechanics** by P.K.Srivastava-NewAge International.
- 14) **Mathematical Physics** by SatyaPrakash- Sultan Chand & Sons.
- 15) **Vector Analysis** by Murray R.Spiegel-2nd edition-Schaum's Outlines,Mc GrawHill
Education


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Department of Physics

KAKATIYA UNIVERSITY - WARANGAL - TELANGANA

Under Graduate Courses (Under CBCS 2020 – 2021 onwards)

B.Sc. PHYSICS II Year

SEMESTER – III

PAPER – III: ELECTROMAGNETIC THEORY

Theory:	4 Hours/Week;	Credits: 4	Marks: 100 (Internal: 20; External: 80)
Practical:	3 Hours/Week	Credits: 1	Marks: 25

UNIT I

Electrostatics

Electric Field:- Concept of electric field lines and electric flux, Gauss's law (Integral and differential forms), application to linear, plane and spherical charge distributions, Conservative nature of electric field 'E', Irrotational field. Electric potential: Concept of electric potential, relation between electric potential and electric field, potential energy of a system of charges, Energy density in an electric field, Calculation of potential from electric field for a spherical charge distribution.

UNIT II

Magnetostatics

Concept of magnetic field 'B' and magnetic flux, Biot-Savart's law, 'B' due to a straight current carrying conductor, Force on a point charge in a magnetic field, Properties of B, curl and divergence of B, solenoidal field, Integral form of Ampere's law, Applications of Ampere's law: field due to straight, circular and solenoidal currents. Energy stored in magnetic field. Magnetic energy in terms of current and inductance, Magnetic force between two current carrying conductors, Magnetic field intensity, Ballistic Galvanometer: Torque on a current loop in a uniform magnetic field, working principle of B.G., current and charge sensitivity, electromagnetic damping, critical damping resistance.

UNIT III:

Electromagnetic Induction and Electromagnetic waves

Faraday's laws of induction (differential and integral form), Lenz's law, self and mutual Induction, Continuity equation, modification of Ampere's law, displacement current, Maxwell equations, Maxwell's equations in vacuum and dielectric medium, boundary conditions, plane wave equation: transverse nature of EM waves, velocity of light in vacuum and in medium, Poynting's theorem.

UNIT IV:

Varying and alternating currents

Growth and decay of currents in LR, CR and LCR circuits - Critical damping, Alternating current, relation between current and voltage in pure R, C and L-vector diagrams - Power in ac circuits. LCR series and parallel resonant circuit-Q-factor, AC & DC motors-single phase, three phase (basics only).

Network Theorems

Passive elements, Power sources, Active elements, Network models: T and π Transformations, Superposition theorem, Thevenin's theorem, Norton's theorem, Reciprocity theorem and Maximum power transfer theorem (Simple problems).

Suggested Books:

1. Fundamentals of electricity and magnetism By Arthur F. Kip (McGraw-Hill, 1968)
2. Electricity and magnetism by J. H. Fewkes & John Yarwood. Vol. I (Oxford Univ. Press, 1991).
3. Introduction to Electrodynamics, 3rd edition, by David J. Griffiths, (Benjamin Cummings, 1998).
4. Electricity and magnetism By Edward M. Purcell (McGraw-Hill Education, 1986)
5. Electricity and magnetism. By D C Tayal (Himalaya Publishing House, 1988)
6. Electromagnetics by Joseph A. Edminister 2nd ed. (New Delhi: Tata McGraw Hill, 2006).

**PAPER – III: ELECTROMAGNETIC THEORY
PRACTICALS**

1. To verify the Thevenin's Theorem
2. To verify Norton Theorem
3. To verify Superposition Theorem
4. To verify maximum power transfer theorem.
5. To determine a small resistance by Carey Foster's bridge.
6. To determine the (a) current sensitivity, (b) charge sensitivity, and (c) CDR of a B.G.
7. To determine high resistance by leakage method.
8. To determine the ratio of two capacitances by De Sauty's bridge.
9. To determine self-inductance of a coil by Anderson's bridge using AC.
10. To determine self-inductance of a coil by Rayleigh's method.
11. To determine coefficient of Mutual inductance by absolute method.

Note: Minimum of eight experiments should be performed.

Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books:

1. B. L. Worsnop and H. T. Flint: Advanced Practical Physics, Asia Publishing House, New Delhi.
2. Indu Prakash and Ramakrishna, A Text Book of Practical Physics, Kitab Mahal

PAPER – V:: (A) MODERN PHYSICS

(DSE-I: ELECTIVE)

Theory:	4 Hours/Week;	Credits: 4	Marks: 100 (Internal: 20; External: 80)
Practical:	3 Hours/Week	Credits: 1	Marks: 25

UNIT - I

SPECTROSCOPY

Atomic Spectra: Introduction - Drawbacks of Bohr's atomic model – Sommerfeld's elliptical orbits -relativistic correction (no derivation). Stern & Gerlach experiment, Vector atom model and quantum numbers associated with it. L-S and j-j coupling schemes. Spectral terms, selection rules, intensity rules – spectra of alkali atoms, doublet fine structure, Zeeman Effect, Paschen-Back Effect and Stark Effect (basic idea).

Molecular Spectroscopy: Types of molecular spectra, pure rotational energies and spectrum of diatomic molecule. Determination of inter nuclear distance. Vibrational energies and spectrum of diatomic molecule. Raman effect, classical theory of Raman effect. Experimental arrangement for Raman effect and its applications.

UNIT – II

Quantum Mechanics

Inadequacy of classical Physics: Spectral radiation - Planck's law (only discussion). Photoelectric effect - Einstein's photoelectric equation. Compton's effect - experimental verification.

Matter waves & Uncertainty principle: de Broglie's hypothesis - wavelength of matter waves, properties of matter waves. Phase and group velocities. Davisson and Germer experiment. Double slit experiment. Standing de Broglie waves of electron in Bohr orbits. Heisenberg's uncertainty principle for position and momentum (x and p_x), Energy and time (E and t). Gamma ray microscope. Diffraction by a single slit. Position of electron in a Bohr orbit. Complementary principle of Bohr.

Schrodinger Wave Equation

Schrodinger time independent and time dependent wave equations. Wave function properties - Significance. Basic postulates of quantum mechanics. Operators, eigen functions and eigen values, expectation values.

UNIT - III

Nuclear Physics

Nuclear Structure: Basic properties of nucleus - size, charge, mass, spin, magnetic dipole moment and electric quadrupole moment. Binding energy of nucleus, deuteron binding energy, p-p, n-n, and n-p scattering (concepts), nuclear forces. Nuclear models - liquid drop model, shell model.

Alpha and Beta Decays: Range of alpha particles, Geiger – Nuttall law. Gamow's theory of alpha decay. Geiger – Nuttall law from Gamow's theory. Beta spectrum - neutrino hypothesis,

Particle Detectors: GM counter, proportional counter, scintillation counter.

UNIT:IV

Solid State Physics & Crystallography

Crystal Structure: Crystalline nature of matter, Crystal lattice, Unit Cell, Elements of symmetry. Crystal systems, Bravais lattices. Miller indices. Simple crystal structures (S.C., BCC, FCC, CsCl, NaCl, diamond and ZincBlende)

Mrs. G. Manjula, Chairperson, BoS

Prof. B. Venkatram Reddy, HoD

Department of Physics

X-ray Diffraction: Diffraction of X -rays by crystals, Bragg's law, Experimental techniques - Laue's method and powder method.

Bonding in Crystals: Types of bonding in crystals - characteristics of crystals with different bondings. Lattice energy of ionic crystals- determination of Madelung constant for NaCl crystal, Calculation of Born Coefficient and repulsive exponent. Born-Haber cycle.

Suggested books:

1. Modern Physics by G. Aruldas & P.Rajagopal. Eastern Economy Edition.
2. Concepts of Modern Physics by Arthur Beiser. Tata McGraw-Hill Edition.
3. Modern Physics by R. Murugesan and Kiruthiga SivaPrasath. S. Chand & Co.
4. Nuclear Physics by D.C. Tayal, Himalaya Publishing House.
5. Molecular Structure and Spectroscopy by G. Aruldas. Prentice Hall of India, New Delhi.
6. Spectroscopy -Atomic and Molecular by Gurdeep R Chatwal and Shyam Anand -Himalaya Publishing House.
7. Third Year Physics – Telugu Academy.
8. Elements of Solid State Physics by J.P. Srivastava. (for chapter on nanomaterials)-Prentice-hall of India Pvt. Ltd.



Mrs. G. Manjula, Chairperson, BoS



Prof. B. Venkatram Reddy, HoD

KAKATIYA UNIVERSITY - WARANGAL - TELANGANA

Under Graduate Courses (Under CBCS 2021 – 2022 onwards)

B.Sc. PHYSICS III Year

SEMESTER – V

**PAPER – V:: (A) MODERN PHYSICS PRACTICALS
(DSE-1: ELECTIVE)**

1. Measurement of Planck's constant using black body radiation and photo-detector
2. Photo-electric effect: photo current versus intensity and wavelength of light; maximum energy of photo-electrons versus frequency of light
3. To determine the Planck's constant using LEDs of at least 4 different colors.
4. To determine the ionization potential of mercury.
5. To determine the absorption lines in the rotational spectrum of Iodine vapour.
6. To determine the value of e/m by (a) Magnetic focusing or (b) Bar magnet.
7. To setup the Millikan oil drop apparatus and determine the charge of an electron.
8. To show the tunneling effect in tunnel diode using I-V characteristics.
9. To determine the wavelength of laser source using diffraction of single slit.
10. To determine the wavelength of laser source using diffraction of double slits.
11. To determine (1) wavelength and (2) angular spread of He-Ne laser using plane diffraction grating
12. To determine the value of e/m for electron by long solenoid method.
13. Photo Cell – Determination of Planck's constant.
14. To verify the inverse square law of radiation using a photo-electric cell.
15. To find the value of photo electric work function of a material of the cathode using a photo-electric cell.
16. Measurement of magnetic field – Hall probe method.
17. To determine the dead time of a given G.M. tube using double source.
18. Hydrogen spectrum – Determination of Rydberg's constant
19. Energy gap of intrinsic semi-conductor
20. G. M. Counter – Absorption coefficients of a material.
21. To draw the plateau curve for a Geiger Muller counter.
22. To find the half-life period of a given radioactive substance using a G.M. Counter.

Reference Books:

1. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
2. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
3. A Text Book of Practical Physics, I. Prakash & Ramakrishna, 11th Edn, 2011, Kitab Mahal

Note: Minimum of eight experiments should be performed.



Mrs. G. Manjula, Chairperson, BoS



Prof. B. Venkatram Reddy, HoD

KAKATIYA UNIVERSITY-WARANGAL-TELANGANA

Under Graduate Courses (Under CBCS 2025–2026 onwards)

B.Sc (PHYSICS) I Year, SEMESTER – II

Paper-II: Thermal Physics

w.e.f academic year (2025-26) (CBCS)

Total: 56 Hrs (4hrs/week)

UNIT-I

Kinetic theory of gases: (4 hrs)

Introduction-Deduction of Maxwell's law of distribution of molecular speeds, Transport

Phenomena-Viscosity of gases-thermal conductivity – diffusion of gases.

Thermodynamics: (8 hrs)

Basics of Thermodynamics-Carnot's engine (qualitative)-Carnot's theorem-Kelvin's and

Clausius statements-Thermodynamic scale of temperature-Entropy, physical significance-

Change in entropy in reversible and irreversible processes-Entropy and disorder-Entropy of

universe -Temperature-Entropy (T-S) diagram - Change of entropy of a perfect gas-change of

entropy when ice changes into steam, Application of entropy in waste management.

UNIT-II

Thermodynamic potentials and Maxwell's equations: (8 hrs)

Thermodynamic potentials-Derivation of Maxwell's thermodynamic relations-Clausius-

Clayperon's equation-Derivation for ratio of specific heats-Derivation for difference of two specific heats for perfect gas.

Low temperature Physics: (8 hrs)

Joule Kelvin effect-liquefaction of gas using porous plug experiment, Joule expansion-Distinction between adiabatic and Joule Thomson expansion-Expression for Joule Thomson cooling-Liquefaction of helium, Kapitza's method-Adiabatic demagnetization-Production of low temperatures -Principle of refrigeration, vapour compression type, Thermocouple-seebeck effect, Peltier effect and Thomson's effect.

UNIT-III

Quantum theory of radiation: (14 hrs)

Black body-Ferry's black body-distribution of energy in the spectrum of Black body-

Wein's displacement law, Wein's law, Rayleigh-Jean's law-Quantum theory of radiation-Planck's law-deduction of Wein's law, Rayleigh-Jeans law, Stefan's law from Planck's law.

Measurement of radiation using pyrometers-Disappearing filament optical pyrometer-experimental determination-Angstrom pyro heliometers-determination of solar constant, effective temperature of sun.

UNIT-IV

Statistical Mechanics: (14 hrs)

Introduction, postulates of statistical mechanics. Phase space, concept of ensembles and some known ensembles, classical and quantum statistics and their differences, concept of probability, Maxwell-Boltzmann's distribution law - Molecular energies in an ideal gas - Maxwell-Boltzmann's velocity distribution law (qualitative), Bose-Einstein Distribution law- application to Photon energy, Fermi-Dirac Distribution law- free electron gas, comparison of three distribution laws.

NOTE: Problems should be solved at the end of every chapter of all units.



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Reference/Suggested books

- 1) **Fundamentals of Physics**. by Halliday/Resnick/Walker. *C. Wiley India Edition 2007.*
- 2) **Second Year Physics –Telugu Academy, Telangana**
- 3) **Modern Physics** by R. Murugesan and Kiruthiga Siva Prasath (For Statistical mechanics) S. Chand & Co.
- 4) **Modern Physics** by G. Aruldas and P. Rajagopal, *Eastern Economy Education.*
- 5) **Statistical Physics** by F. Reif Berkeley Physics Course. Volume-5, *The McGraw-Hill Companies.*
- 6) **An Introduction to Thermal Physics** by Daniel V. Schroeder. *Pearson Education Low Price Edition.*
- 7) **Thermodynamics** by R.C. Srivastava, Subit K. Saha & Abhay K. Jain *Eastern Economy Edition.*
- 8) **Modern Engineering Physics** by A.S. Vasudeva *.S. Chand & Co. Publications.*
- 9) **Feynman's Lectures on Physics** Vol. 1, 2, 3 & 4. *Narosa Publications.*
- 10) **Heat and Thermo dynamics:** K.W. Zeemansky.
- 11) **Introduction to statistical Mechanics** by B.B. Laud (Macmillan 1981).
- 12) **Statistical Physics** by F. Reif., (Mc Graw-Hill, 1998)
- 13) **Statistical Physics** by K. Huang., (Wiley Eastern 1988)

B.Sc (Physics) - I year, Semester - II
Paper-II: Thermal Physics Practical lab

No. of hours per week: 2

- 1) Determination of Co-efficient of thermal conductivity of a bad conductor by Lee's method.
- 2) Determination of Stefan's constant-Stefan's experiment.
- 3) Determination of Specific heat of a liquid by using Newton's law of cooling method.
- 4) Determination of heating efficiency of electrical kettle with varying voltages.
- 5) Cooling Curve of a metallic body (Null method).
- 6) Determination of temperature coefficient of resistance using resistance thermometer.
- 7) Study of conversion of mechanical energy to heat.
- 8) Determination of Specific heat of a solid (graphite).
- 9) Thermal expansion of solids
- 10) Calibration of thermo couple
- 11) Simulations for T-S diagram

Note: Minimum of **Eight** experiments should be performed. Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books

- 1) D.P.Khandelwal, "A laboratory manual for under graduate classes"(Vani Publishing House, New Delhi).
- 2) S.P.Singh, "Advanced Practical Physics"(Pragati Prakashan, Meerut).
- 3) Worsnop and Flint-Advanced Practical physics for students.
- 4) "Practical Physics" R.KShukla, Anchal Srivastava.


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Department of Physics

KAKATIYA UNIVERSITY - WARANGAL - TELANGANA

Under Graduate Courses (Under CBCS 2020 – 2021 onwards)

B.Sc. PHYSICS II Year

SEMESTER – IV

PAPER – IV:: WAVES AND OPTICS

Theory:	4 Hours/Week;	Credits: 4	Marks: 100 (Internal: 20; External: 80)
Practical:	3 Hours/Week	Credits: 1	Marks: 25

UNIT-I:

Waves

Fundamentals of Waves -Transverse wave propagation along a stretched string, general solution of wave equation and its significance, modes of vibration of stretched string clamped at ends, overtones, energy transport, transverse impedance.

Longitudinal vibrations in bars- wave equation and its general solution, Special cases: (i) bar fixed at both ends, ii) bar fixed at the midpoint, iii) bar free at both ends, iv) bar fixed at one end, Transverse vibrations in a bar - wave equation and its general solution. Boundary conditions, clamped free bar, free-free bar, bar supported at both ends, Tuning fork.

UNIT II:

Interference

Principle of superposition – coherence – temporal coherence and spatial coherence – conditions for Interference of light.

Interference by division of wave front: Fresnel's biprism – determination of wave length of light. Determination of thickness of a transparent material using biprism – change of phase on reflection – Lloyd's mirror experiment.

Interference by division of amplitude: Oblique incidence of a plane wave on a thin film due to reflected and transmitted light (Cosine law) – Colours of thin films – Non-reflecting films – interference by a plane parallel film illuminated by a point source – Interference by a film with two non-parallel reflecting surfaces (Wedge shaped film) – Determination of diameter of wire-Newton's rings in reflected light with and without contact between lens and glass plate, Newton's rings in transmitted light (Haidinger Fringes) – Determination of wave length of monochromatic light – Michelson Interferometer – types of fringes – Determination of wavelength of monochromatic light, Difference in wavelength of sodium D_1, D_2 lines and thickness of a thin transparent plate.

UNIT III:

Diffraction:

Introduction – Distinction between Fresnel and Fraunhofer diffraction, Fraunhofer diffraction:- Diffraction due to single slit and circular aperture – Limit of resolution – Fraunhofer diffraction due to double slit – Fraunhofer diffraction pattern with N slits (diffraction grating).

Resolving Power of grating – Determination of wave length of light in normal and oblique incidence methods using diffraction grating.

Fresnel diffraction-Fresnel's half period zones – area of the half period zones –zone plate – Comparison of zone plate with convex lens – Phase reversal zone plate – diffraction at a straight edge – difference between interference and diffraction.

UNIT IV:

Polarization

Polarized light : Methods of Polarization, Polarization by reflection, refraction, Double refraction, selective absorption , scattering of light – Brewster's law – Malus law – Nicol prism polarizer and analyzer – Refraction of plane wave incident on negative and positive crystals (Huygen's explanation) – Quarter wave plate, Half wave plate – Babinet's compensator – Optical activity, analysis of light by Laurent's half shade polarimeter.

NOTE: Problems should be solved at the end of every chapter of all units.

Suggested books

1. **Optics** by Ajoy Ghatak. *The McGraw-Hill companies.*
2. **Optics** by Subramaniam and Brijlal. *S. Chand & Co.*
3. **Fundamentals of Physics.** Halliday/Resnick/Walker. *C. Wiley India Edition 2007.*
4. **Optics and Spectroscopy.** R. Murugesan and Kiruthiga Siva Prasath. *S. Chand & Co.*
5. **Second Year Physics – Telugu Academy.**
1. **Modern Engineering Physics** by A.S. Vasudeva. *S.Chand & Co. Publications.*
2. **Feynman's Lectures on Physics** Vol. 1, 2, 3 & 4. *Narosa Publications.*
3. **Fundamentals of Optics** by Jenkins A. Francis and White E. Harvey, *McGraw Hill Inc.*
4. K. Ghatak, **Physical Optics'**
5. D.P. Khandelwal, **Optical and Atomic Physics'** (Himalaya Publishing House, Bombay, 1988)
11. Jenkins and White: **'Fundamental of Optics'** (McGraw-Hill)
12. Smith and Thomson: **'Optics'** (John Wiley and sons).



Department of Physics

KAKATIYA UNIVERSITY - WARANGAL - TELANGANA

Under Graduate Courses (Under CBCS 2020 – 2021 onwards)

B.Sc. PHYSICS II Year

SEMESTER – IV

PAPER – IV:: WAVES AND OPTICS
PRACTICALS

1. Thickness of a wire using wedge method.
2. Determination of wavelength of light using Biprism.
3. Determination of Radius of curvature of a given convex lens by forming Newton's rings.
4. Resolving power of grating.
5. Study of optical rotation- polarimeter.
6. Dispersive power of a prism
7. Determination of wavelength of light using diffraction grating minimum deviation method.
8. Wavelength of light using diffraction grating – normal incidence method.
9. Resolving power of a telescope.
10. Refractive index of a liquid and glass (Boys Method).
11. Pulfrich refractometer – determination of refractive index of liquid.
12. Wavelength of Laser light using diffraction grating.
13. Verification of Laws of a stretched string (Three Laws).
14. Velocity of Transverse wave along a stretched string
15. Determination of frequency of a bar- Melde's experiment

Note: Minimum of eight experiments should be performed Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books

1. D.P. Khandelwal, "A laboratory manual for undergraduate classes" (Vani Publishing House, New Delhi).
2. S.P. Singh, "Advanced Practical Physics" (Pragati Prakashan, Meerut).
3. Worsnop and Flint- Advanced Practical physics for students.
4. "Practical Physics" R.K Shukla, Anchal Srivastav.

PAPER – VI :: (A) ELECTRONICS

(DSE-2: ELECTIVE)

Theory:	4 Hours/Week;	Credits: 4	Marks: 100 (Internal: 20; External: 80)
Practical:	3 Hours/Week	Credits: 1	Marks: 25

Unit - I

Band theory of P-N junction: Energy band in solids (band theory), valence band, conduction band and forbidden energy gap in solids, insulators, semi conductors and pure or intrinsic semiconductors and impure or extrinsic semi-conductors. N-type semi-conductors, P-type semi-conductors, Fermi level, continuity equation.

Diodes: P-N junction diode, Half-wave, full-wave and bridge rectifier. Zener diode & its characteristics. Zener diode as voltage regulator.

UNIT-II

Bipolar Junction Transistor (BJT) – p-n-p and n-p-n transistors, current components in transistors, CB, CE and CC configurations – transistor as an amplifier -RC coupled amplifier – Frequency response (Qualitative analysis).

Feedback concept & Oscillators: Feedback, General theory of feedback–Concepts of oscillators, Barkhausen's criteria, Phase shift oscillator – Expression for frequency of oscillation.

UNIT-III

Special devices- Construction and Characteristics: Photo diode - Shockley diode -Solar cell, Opto-couplers - Field Effect Transistor (FET) - FET as an Amplifier - Uni Junction Transistor (UJT), UJT as a relaxation oscillator - Silicon controlled rectifier (SCR) - SCR as a switch.

UNIT-IV

Digital Electronics

Binary number system, conversion of binary to decimal and vice-versa. Binary addition and subtraction (1's and 2's complement methods). Hexadecimal number system. Conversion from binary to hexadecimal and vice-versa, Decimal to hexadecimal and vice-versa.

Logic gates:

OR, AND, NOT gates, truth tables, realization of these gates using discrete components. NAND, NOR as universal gates, Exclusive – OR gate (EX-OR). De Morgan's Laws – Verification.

NOTE: Problems should be solved from every chapter of all units.

Suggested Books:

1. Electronic devices and circuits – Millman and Halkias. *Mc.Graw-Hill Education*.
2. Principles of Electronics by V.K. Mehta – *S. Chand & Co.*
3. Basic Electronics (Solid state) – B. L. Theraja, S. Chand & Co.
4. A First Course in Electronics- Anwar A. Khan & Kanchan K. Dey, PHI.
5. Physics of Semiconductor Devices- S. M. Sze
6. Physics of Semiconductors- Streetman.
7. Basic Electronics – Bernod Grob.
8. Basic Electronics for B.Sc (Physics) III Year, 2019, Telugu Academy
9. Digital Principles & Applications – A.P. Malvino and D.P. Leach

PAPER – VI:: (A) ELECTRONICS PRACTICALS

(DSE-2: ELECTIVE)

1. Construction of logic gates (AND, OR, NOT, gates) with discrete components– Truth table Verification
2. AND, OR, NOT – gates constructions using universal gates – Verification of truth tables.
3. Construction of NAND and NOR gates with discrete components and truth table verification
4. Characteristics of a Transistor in CE configuration
5. R.C. coupled amplifier – frequency response.
6. Verification of De Morgan's Theorem.
7. Zener diode V-I characteristics.
8. P-n junction diode V- I characteristics.
9. Zener diode as a voltage regulator
10. Construction of a model D.C. power supply
11. R C phase shift Oscillator –determination of output frequency

Note: Minimum of eight experiments should be performed.

Suggested Books:

1. B.Sc. Practical Physics – C. L. Arora – S. Chand & Co.
2. Viva-voce in Physics – R.C. Gupta, Pragathi Prakashan, Meerut.
3. Laboratory manual for Physics Course by B.P. Khandelwal.
4. Practical Physics by M. Arul Thakpathi by Comptex Publishers.
5. B.Sc. practical physics – Subbi Reddy.

Time table (A.Y.: 2026-2027)

	1	2	3	4	5	6	7
Mon			WRM	MPCs-III		LAB: MPCs-I (B-I)	
Tue	MPCs-I		RC: MPCs-II	MPCs-III	MPCs-II	RC: MPCs-III	
Wed	MPCs-II	WRM			MPCs-I	LAB: MPCs-I (B-I)	
Thu	MPCs-III	MPCs-II				LAB: MPCs-II	
Fri	MPCs-II	MPCs-I			LAB: MPCs-III		
Sat	MPCs-III		MPCs-I			RC: MPCs-I	

Future plans:

- ✓ Introduce project-based learning and industry-relevant topics (e.g., Photonics, Materials Science, Nanotechnology).
- ✓ Establish a departmental research cell to encourage student and faculty research.
- ✓ Promote mini-projects, undergraduate research, and collaborations with local industries, research institutes, and universities.
- ✓ Upgrade existing laboratories with modern equipment and ICT-enabled experiments.